Statistics

Chapter 7 — Instructor Solutions Manual

October 12, 2026 • Total Exercises: 7

PROBLEM STATEMENT

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Step 1

DEFINITIONS

General multiplication rule:

$$P(A \text{ and } B) = P(A \cap B) = P(A) \times P(B|A) = P(B) \times P(A|B)$$

Complement rule:

$$P(\sim A) = P(\text{not } A) = 1 - P(A)$$

Step 2

SOLUTION

(a) 5 of the 20 tax returns are business.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{First business}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{5}{20}$$

Once one of the business tax returns have been selected, 4 of the 19 remaining tax returns are business.

$$P(\text{Second business}|\text{First business}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{4}{19}$$

Use the general multiplication rule:

$$\begin{aligned} P(\text{both business}) &= P(\text{First business}) \\ &\quad \times P(\text{Second business}|\text{First business}) \\ &= \frac{5}{20} \times \frac{4}{19} \\ &= \frac{1}{4} \times \frac{4}{19} \\ &= \frac{1 \times 4}{4 \times 19} \\ &= \frac{1}{19} \\ &\approx 0.0526 \\ &= 5.26\% \end{aligned}$$

Step 3

(b) 5 of the 20 tax returns are business, thus $20 - 5 = 15$ of the 20 tax returns are not business.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{First not business}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{15}{20}$$

Once one of the business tax returns have been selected, 4 of the 19 remaining tax returns are business.

$$P(\text{Second not business}|\text{First not business}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{14}{19}$$

Use the general multiplication rule:

$$\begin{aligned} P(\text{none business}) &= P(\text{First not business}) \\ &\quad \times P(\text{Second not business}|\text{First not business}) \\ &= \frac{15}{20} \times \frac{14}{19} \\ &= \frac{3}{4} \times \frac{14}{19} \\ &= \frac{3 \times 14}{4 \times 19} \\ &= \frac{3 \times 7}{2 \times 19} \\ &= \frac{21}{38} \end{aligned}$$

Use the complement rule:

$$\begin{aligned} P(\text{at least one business}) &= 1 - P(\text{none business}) \\ &= 1 - \frac{21}{38} \\ &= \frac{17}{38} \\ &\approx 0.4474 \\ &= 44.74\% \end{aligned}$$

Result

(a) $\frac{1}{19} \approx 0.0526 = 5.26\%$

(b) $\frac{17}{38} \approx 0.4474 = 44.74\%$

PROBLEM STATEMENT

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Step 1

DEFINITIONS

Definition **binomial probability**:

$$P(X=k) =_n C_k \cdot \pi^k \cdot (1-\pi)^{n-k} = \frac{n!}{k!(n-k)!} \cdot \pi^k \cdot (1-\pi)^{n-k}$$

Complement rule:

$$P(\sim A) = P(\text{not } A) = 1 - P(A)$$

Step 2

SOLUTION

$$n = \text{Sample size} = 16$$
$$\pi = \text{Probability of success} = 15\% = 0.15$$

(a) Evaluate the definition of binomial probability at $k = 1$:

$$\begin{aligned} P(X=1) &=_{16} C_1 \cdot 0.15^1 \cdot (1-0.15)^{16-1} \\ &= \frac{16!}{1!(16-1)!} \cdot 0.15 \cdot 0.85^{15} \\ &= 16 \cdot 0.15 \cdot 0.85^{15} \\ &\approx 0.2097 \\ &= 20.97\% \end{aligned}$$

Step 3

(b) Evaluate the definition of binomial probability at $k = 0$:

$$\begin{aligned} P(X = 0) &= {}_{16}C_0 \cdot 0.15^0 \cdot (1 - 0.15)^{16-0} \\ &= \frac{16!}{0!(16-0)!} \cdot 1 \cdot 0.85^{16} \\ &= 1 \cdot 1 \cdot 0.85^{16} \\ &= 0.85^{16} \\ &\approx 0.0743 \end{aligned}$$

Use the complement rule:

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) \\ &= 1 - 0.0743 \\ &= 0.9257 \\ &= 92.57\% \end{aligned}$$

Result

(a) $0.2097 = 20.97\%$

(b) $0.9257 = 92.57\%$

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Step 1

Fundamental counting principle (multiplication rule): If the first event could occur in m ways and the second event could occur in n ways, then the number of ways that the two events could occur in sequence is $m \cdot n$.

There are 6 possible CPAs (Fred and five others), while there are 5 parking spots for the 6 CPAs.

First parking spot: 6 ways

Second parking spot: 5 ways $\text{\color{#4257b2}(different from CPA in previous parking spot)\color{default}

Third parking spot: 4 ways \color{#4257b2}(different from CPA in previous parking spots)\color{default}

Fourth parking spot: 3 ways \color{#4257b2}(different from CPA in previous parking spots)\color{default}

Fifth parking spot: 2 ways \color{#4257b2}(different from CPA in previous parking spots)\color{default}\

Use the fundamental counting principle:

$$6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 = 720$$

Thus there are 720 different ways the cars can be arranged in the five parking spots.}$

Result

720 different ways

PROBLEM STATEMENT

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Step 1

Let x be the number of exemptions per return and $P(x)$ be the percent of the people who availed exemptions.

a.The mean number of exemptions per return

Mean number of exemptions per return is μ.

$$\begin{aligned}\mu &= \Sigma x P(x)\\ &= 1 \times \frac{20}{100} + 2 \times \frac{50}{100} + 3 \times \frac{20}{100} + 4 \times \frac{10}{100}\\ &= 0.2 + 2 \times 0.5 + 3 \times 0.2 + 4 \times 0.10\\ &= 0.2 + 1 + 0.6 + 0.4\\ &= 2.2\end{aligned}$$

Step 2

b.The variance of the number of exemptions per return

Variance of the number of exemptions per return is σ^2.

$$\begin{aligned}\sigma^2 &= \Sigma x^2 P(x) - \mu^2\\ &= 1^2 \times \frac{20}{100} + 2^2 \times \frac{50}{100} + 3^2 \times \frac{20}{100} + 4^2 \times \frac{10}{100} - (2.2)^2\\ &= 1 \times 0.2 + 4 \times 0.5 + 9 \times 0.2 + 16 \times 0.1 - 4.84\\ &= 0.2 + 2 + 1.8 + 1.6 - 4.84\\ &= 5.6 - 4.84\\ &= 0.76\end{aligned}$$

Result

$$\mu = 2.2$$

$$\sigma^2 = 0.76$$

PROBLEM STATEMENT

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Step 1

Mean amount of the refund is $\mu = \$1600$ and the standard deviation is $\sigma = \$850$

Let X be the refunds.

$\textbf{a. X is between 1600 and 2000 }$

$$\begin{aligned} P(1600 < X < 2000) &= P(1600 - \mu < X - \mu < 2000 - \mu) \\ &= P\left(\frac{1600 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{2000 - \mu}{\sigma}\right) \\ &= P\left(\frac{1600 - 1600}{850} < z < \frac{2000 - 1600}{850}\right) \\ &= P\left(\frac{0}{850} < z < \frac{400}{850}\right) \\ &= P(0 < z < 0.47) \\ &= 0.1808 \ \ (\text{from Appendix B1.}) \end{aligned}$$

18.08% of the refunds are between $\$1600$ and $\$2000$.

Step 2

$\textbf{b. } X \text{ is between } 900 \text{ and } 2000 \text{ }}$

$$\begin{aligned} P(900 < X < 2000) &= P(900 - \mu < X - \mu < 2000 - \mu) \\ &= P\left(\frac{900 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{2000 - \mu}{\sigma}\right) \\ &= P\left(\frac{900 - 1600}{850} < z < \frac{2000 - 1600}{850}\right) \\ &= P\left(\frac{-700}{850} < z < \frac{400}{850}\right) \\ &= P(-0.82 < z < 0.47) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-0.82 < z < 0)$, $P(0 < z < 0.47)$ and add the two probabilities.

$$\begin{aligned} P(900 < X < 2000) &= P(-0.82 < z < 0) + P(0 < z < 0.47) \\ &= P(0 < z < 0.82) + P(0 < z < 0.47) \\ &= 0.2939 + 0.1808 \ \ (\text{from Appendix B1.}) \\ &= 0.4747 \end{aligned}$$

47.47% of the refunds are between $\$900$ and $\$2000$.

Step 3

$\textbf{c. } X \text{ is less than } 0 \text{ }}$

$$\begin{aligned} P(X < 0) &= P(X - \mu < 0 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} < \frac{0 - \mu}{\sigma}\right) \\ &= P\left(z < \frac{0 - 1600}{850}\right) \\ &= P\left(z < \frac{-1600}{850}\right) \\ &= P(z < -1.88) \\ &= P(z > 1.88) \\ &= 0.5 - P(0 < z < 1.88) \\ &= 0.5 - 0.4699 \ \ (\text{from Appendix B1.}) \\ &= 0.0301 \end{aligned}$$

3.01% of the funds are less than 0.

🎯 **Result**

a. 18.08%

b. 47.47%

c. 3.01%

PROBLEM STATEMENT

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Step 1

Given:

Refund	1	2	3 or more	Total
Yes	20	20	10	50
No	10	20	0	30
Total	30	40	10	80

(a) A table that contains counts for two different (qualitative) variables is called a **contingency table**.

Step 2

(b) The table describes 80 returns in total.

50 of the 80 returns result in a refund, because 50 is mentioned in the row "Yes" and in the column "Total" of the given table.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{refund}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{50}{80} = \frac{5}{8} = 0.625 = 62.5\%$$

Step 3

(c) The table describes 80 returns in total.

50 of the 80 returns resulted in a refund, because 50 is mentioned in the row "Yes" and in the column "Total" of the given table.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{refund}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{50}{80}$$

30 of the 80 returns had 1 dependent, because 30 is mentioned in the row "Total" and in the column "1" of the given table.

$$P(1\text{ dependent}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{30}{80}$$

20 of the 80 returns resulted in a refund and had 1 dependent, because 20 is mentioned in the row "Yes" and in the column "1" of the given table.

$$P(\text{refund and 1 dependent}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{20}{80}$$

Use the **General addition rule** for any two events: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$$\begin{aligned}
&P(\text{refund or 1 dependent}) \\
&= P(\text{refund}) + P(1\text{ dependent}) - P(\text{refund and 1 dependent}) \\
&= \frac{50}{80} + \frac{30}{80} - \frac{20}{80} \\
&= \frac{50 + 30 - 20}{80} \\
&= \frac{60}{80} \\
&= \frac{3}{4} \\
&= 0.75 \\
&= 75\%
\end{aligned}$$

Step 4

(d) The table describes 80 returns in total.

50 of the 80 returns resulted in a refund, because 50 is mentioned in the row "Yes" and in the column "Total" of the given table.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{refund}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{50}{80}$$

20 of the 80 returns resulted in a refund and had 1 dependent, because 20 is mentioned in the row "Yes" and in the column "1" of the given table.

$$P(\text{refund and 1 dependent}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{20}{80}$$

Use the Definition **Conditional probability**: $P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A \text{ and } B)}{P(A)}$

$$\begin{aligned} P(1 \text{ dependent}|\text{refund}) &= \frac{P(\text{refund and 1 dependent})}{P(\text{refund})} \\ &= \frac{20/80}{50/80} \\ &= \frac{20}{50} \\ &= \frac{2}{5} \\ &= 0.4 \\ &= 40\% \end{aligned}$$

Step 5

(e) The table describes 80 returns in total.

10 of the 80 returns resulted in no refund and had 1 dependent, because 10 is mentioned in the row "No" and in the column "1" of the given table.

$$\begin{aligned} P(\text{no refund and 1 dependent}) &= \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} \\ &= \frac{10}{80} \\ &= \frac{1}{8} \\ &= 0.125 \\ &= 12.5\% \end{aligned}$$

Result

(a) Contingency table

(b) $\frac{5}{8} = 0.625 = 62.5\%$

(c) $\frac{3}{4} = 0.75 = 75\%$

(d) $\frac{2}{5} = 0.4 = 40\%$

(e) $\frac{1}{8} = 0.125 = 12.5\%$

PROBLEM STATEMENT

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Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X = k) = \frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A) = P(\text{not } A) = 1 - P(A)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A \cup B) = P(A \text{ or } B) = P(A) + P(B)$$

Multiplication rule for independent events:

$$P(A \cap B) = P(A \text{ and } B) = P(A) \times P(B)$$

Step 2

SOLUTION

$$\mu = \text{Mean} = 3$$

(a) Evaluate the probability of Poisson probability at $x = 0$:

$$\begin{aligned} P(X = 0) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{3^0 e^{-3}}{0!} \\ &= \frac{1 e^{-3}}{1} \\ &= e^{-3} \\ &\approx 0.0498 \\ &= 4.98\% \end{aligned}$$

Step 3

(b) Evaluate the probability of Poisson probability at $x = 3$:

$$\begin{aligned} P(X = 3) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{3^3 e^{-3}}{3!} \\ &= \frac{27e^{-3}}{6} \\ &= \frac{9}{2}e^{-3} \\ &\approx 0.2240 \\ &= 22.40\% \end{aligned}$$

Step 4

(c) Evaluate the probability of Poisson probability at $x = 1, 2, 4$:

$$\begin{aligned} P(X = 1) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{3^1 e^{-3}}{1!} \\ &= \frac{3e^{-3}}{1} \\ &= 3e^{-3} \\ &\approx 0.1494 \\ P(X = 2) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{3^2 e^{-3}}{2!} \\ &= \frac{9e^{-3}}{2} \\ &= \frac{9}{2}e^{-3} \\ &\approx 0.2240 \\ P(X = 4) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{3^4 e^{-3}}{4!} \\ &= \frac{81e^{-3}}{24} \\ &= \frac{27}{8}e^{-3} \\ &\approx 0.1680 \end{aligned}$$

Step 5

Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of successes for the same month):

$$\begin{aligned} P(X<5) &= P(X=0)+P(X=1)+P(X=2) \\ &\quad +P(X=3)+P(X=4) \\ &= 0.0498+0.1494+0.2240 \\ &\quad +0.2240+0.1680 \\ &= 0.8152 \end{aligned}$$

Use the complement rule:

$$\begin{aligned} P(X\geq 5) &= 1-P(X<5) \\ &= 1-0.8152 \\ &= 0.1848 \\ &= 18.45\% \end{aligned}$$

Step 6

(d) Use the multiplication rule for independent events (assuming that the number of requests on a single day is not affected by the number of requests on the previous days) and the results of part (a):

$$\begin{aligned} P(X=0 \text{ for two days}) &= P(X=0)\times P(X=0) \\ &= 0.0498\times 0.0498 \\ &= 0.0498^2 \\ &\approx 0.0025 \\ &= 0.25\% \end{aligned}$$

Result

(a) $e^{-3}\approx 0.0498 = 4.98\%$

(b) $\frac{9}{2}e^{-3}\approx 0.2240 = 22.40\%$

(c) $0.1848 = 18.45\%$

(d) $0.0025 = 0.25\%$